MCQ solutions

1. **D.** The argument must be strictly positive: 2 $x - 1 > 0$. Zero is excluded.
2. **A.** The logarithm turns a product into a sum; no similar rule turns a sum of arguments.
3. **C.** The exponential and logarithm functions are inverses of each other on their domains.
4. **B.** Apply u'/u with $u = 1 + x^2 > 0$; the factor 2 x comes from the composite function.
5. **D.** Apply the exponential to both sides; $e^2 > 0$ satisfies the domain condition.
6. **C.** $2^{-3} = 0{,}125 \geq 0{,}1$ and $2^{-4} = 0{,}0625 < 0{,}1$. The sequence decreases, so no earlier index works.
7. **A.** Take the logarithm then divide by ln $q < 0$ reverses the direction of the inequality.
8. **B.** Comparative growth of the logarithm and a positive power of x.
9. **D.** The right-hand side requires $x > 0$. For $x \neq 0$, the general identity is ln $(x^2) = 2\ln|x|$.
10. **C.** $g' = (1 - \ln x)/x^2$, positive before e then negative. The maximum reached at e is equal to $1/e$.

True/false. False: ln 2 $\neq 0$. False: ln 2 > 0. False: the maximum 1/ $e < 1$ (question 10) rules it out.

Solutions to the applications and guidance

Application 1. Domains: $]-\infty;2[\ ;\ ]-\infty;-1[\cup]1;+\infty[$. For the equation, $x>-1/2$ and $2x+1=1$, hence $x=0$.

Application 2. Take $a=b=1$. For $x=-1$, the left-hand side $\ln 1$ exists but $\ln(-1)$ is not in the reals. The correct identity is $\ln(x^2)=2\ln|x|$ for $x\neq 0$.

Application 3. Domain $\mathbb{R}$, derivative $2x/(1+x^2)$, negative before 0 and positive after; minimum $g(0)=0$, therefore $g\geq 0$. $g(-x)=g(x)$; symmetry confirms the opposite variations on either side of 0.

Application 4. First threshold: $n\geq \ln 2/\ln 1{,}05\simeq 14{,}21$, so 15; $1{,}05^{14}\simeq 1{,}980$ and $1{,}05^{15}\simeq 2{,}079$. Second: $n\geq \ln 0{,}2/\ln 0{,}9\simeq 15.28$, so 16; $0.9^{15}\simeq 0.2059$ and $0.9^{16}\simeq 0{,}1853$.

Application 5. $\ln(2x)/x=\ln 2/x+\ln x/x\to 0$. $x\ln(3x)=x\ln 3+x\ln x\to 0$.

Application 6. $f(x)=\ln x/x$ tends to $-\infty$ at 0^+, strictly increases up to $1/e$, then decreases strictly towards 0 while remaining positive after e. Thus: one solution for $a<0$; one for $a=0$ (1) ; two for $0<a<1/e$; one for $a=1/e$ (e); none for $a>1/e$. The conclusions use the continuity on each branch.

Learning from mistakes

For the threshold, ask pupils first to predict whether the sequence increases or decreases, then to compare this direction with the inequality obtained. Have students revisit the false distributivity with a product in which one factor equals 1. The parameter problem allows work on existence, uniqueness on each branch and limits not attained.

Teaching notes and solutions

Guidance for the teacher

S44 is placed before the first mock exam, but the logarithm is not within its stated scope. Insist on the conditions $a > 0, b > 0$ before combining logarithms. The differentiability of ln is accepted, the differentiation formula is proved. For a threshold with $0 < q < 1$, the neg. sign of ln q reverses the inequality. Have the candidate index and the previous one checked.

Solutions to the activity

1. Each step multiplies the error by 0.7: $e_n = 5(0{,}7)^n$ for $n \geq 0$.

2. $e_n \leq 10^{-4}$ is equivalent to $(0{,}7)^n \leq 2 \times 10^{-5}$. Since ln is increasing, $n \ln 0{,}7 \leq \ln(2 \times 10^{-5})$. Since $\ln 0{,}7 < 0$, we get
$$n \geq \frac{\ln(2 \times 10^{-5})}{\ln 0{,}7} \simeq 30{,}33512.$$
The first suitable integer is 31.

3. $e_{30} \simeq 1{,}1270 \times 10^{-4} > 10^{-4}$ & $e_{31} \simeq 7{,}8888 \times 10^{-5} < 10^{-4}$. The sequence of errors decreases, so this check confirms minimality.

4. For the target 10^{-6}, the same calculation gives $n \geq \frac{\ln(2\times10^{-7})}{\ln 0{,}7} \simeq 43{,}2465$, so $n = 44$. The true threshold increases by $\frac{\ln 100}{|\ln 0{,}7|} \simeq 12{,}91$, then round up to the next integer. The number of steps is not multiplied by 100.

E1. We need $2x - 1 > 0$ and $x + 3 > 0$, therefore $x > \frac{1}{2}$. On this domain, the injectivity of ln gives $2x - 1 = x + 3$, i.e. $x = 4$. This value lies in the domain; the solution set is $\{4\}$.

E2. We need $x > 2$. The product formula gives $\ln(x(x-2)) = \ln 3$, so $x^2 - 2x - 3 = 0$, i.e. $(x-3)(x+1) = 0$. Only $x = 3$ satisfies $x > 2$: the other root -1 is excluded.

E3. Since $x^2 + 1 > 0$, the first function is differentiable on $\mathbb{R}$ and its derivative is $\frac{2x}{x^2+1}$. On $]0, +\infty[$, the product rule gives $(x \ln x)' = \ln x + x(\frac{1}{x}) = 1 + \ln x$.

E4. For the first limit, let $t = \sqrt{x} \to +\infty : \frac{\ln x}{\sqrt{x}} = \frac{2 \ln t}{t} \to 0$. For the second, let $t = -\ln x \to +\infty : x^2 \ln x = -te^{-2t} \to 0$. Both limits use comparative growth.

E5. For $x > 0$, let $g(x) = x - 1 - \ln x$. We have $g'(x) = \frac{x-1}{x}$, negative before 1, positive after. Its minimum is $g(1) = 0$, therefore $g(x) \geq 0$, which is equivalent to $\ln x \leq x - 1$.

Solutions to homework DM6

1. At 0^+, the numerator tends to $-\infty$ and the denominator to $0^+ : f \to -\infty$. At $+\infty$, $f = \frac{1}{x} + \frac{\ln x}{x} \to 0$. The sign is that of $1 + \ln x$: negative on $]0, e^{-1}[$, zero at e^{-1}, positive afterwards. As a quotient,
$$f'(x) = \frac{1 - (1 + \ln x)}{x^2} = -\frac{\ln x}{x^2}.$$
Thus f strictly increases up to 1, then strictly decreases; its maximum is $f(1) = 1$.

2. Differentiating $-(\ln x)x^{-2}$, we get $f''(x) = \frac{2 \ln x - 1}{x^3}$. The numerator changes sign, from negative to positive, at $x = \sqrt{e}$. Hence f is concave on $]0, \sqrt{e}]$, convex on $[\sqrt{e}, +\infty[$ and has the inflection $(\sqrt{e}, \frac{3}{2\sqrt{e}})$.

3. On $]0,1]$, continuity, strict increase, limit $-\infty$ and final value 1 give exactly one preimage of $\frac{1}{2}$. On $[1,+\infty[$, strict decrease from 1 to 0 gives exactly a second preimage. The first is in $]e^{-1},\frac{1}{2}[$, since $f(e^{-1})=0$ and $f(\frac{1}{2})=2(1-\ln 2)\simeq 0{,}6137>\frac{1}{2}$. The second is in $]5,6[$, since $f(5)\simeq 0{,}5219>\frac{1}{2}>f(6)\simeq 0{,}4653$.

4. The maximum gives $\frac{1+\ln x}{x}\leq 1$. Since $x>0$, we can multiply without changing the direction: $1+\ln x\leq x$. Equality corresponds to the unique maximum of f, hence to $x=1$.

Solutions to the textbook or past-paper work

The logarithm requires a strictly positive argument.

1. $3x-7>0\iff x>\frac{7}{3}$, therefore $D_f=]\frac{7}{3};+\infty[$.

2. $-x^2+4x-3=-(x-1)(x-3)>0$ exactly when $1<x<3$. Thus $D_g=]1;3[$.

3. We need both $x>0$ and $2-x>0$; their intersection gives $D_h=]0;2[$. The coefficient -3 placed in front of the second logarithm does not change its domain of definition.